# PG No. 35 $C_{3\mathrm{v}}(1)$ $3m$ (31m setting) [ trigonal ]

Table 1: Representation matrices

| Irrep. | | | | | | |
|---|---|---|---|---|---|---|
| $A_1$ | $1:\begin{bmatrix}1\end{bmatrix}$ | $3^+_{001}:\begin{bmatrix}1\end{bmatrix}$ | $3^-_{001}:\begin{bmatrix}1\end{bmatrix}$ | $\mathrm{m}_{1-10}:\begin{bmatrix}1\end{bmatrix}$ | $\mathrm{m}_{120}:\begin{bmatrix}1\end{bmatrix}$ | $\mathrm{m}_{210}:\begin{bmatrix}1\end{bmatrix}$ |
| $A_2$ | $1:\begin{bmatrix}1\end{bmatrix}$ | $3^+_{001}:\begin{bmatrix}1\end{bmatrix}$ | $3^-_{001}:\begin{bmatrix}1\end{bmatrix}$ | $\mathrm{m}_{1-10}:\begin{bmatrix}-1\end{bmatrix}$ | $\mathrm{m}_{120}:\begin{bmatrix}-1\end{bmatrix}$ | $\mathrm{m}_{210}:\begin{bmatrix}-1\end{bmatrix}$ |
| $E$ | $1:\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ | $3^+_{001}:\begin{bmatrix}-\frac{1}{2} & -\frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & -\frac{1}{2}\end{bmatrix}$ | $3^-_{001}:\begin{bmatrix}-\frac{1}{2} & \frac{\sqrt{3}}{2}\\ -\frac{\sqrt{3}}{2} & -\frac{1}{2}\end{bmatrix}$ | $\mathrm{m}_{1-10}:\begin{bmatrix}-\frac{1}{2} & \frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & \frac{1}{2}\end{bmatrix}$ | $\mathrm{m}_{120}:\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$ | $\mathrm{m}_{210}:\begin{bmatrix}-\frac{1}{2} & -\frac{\sqrt{3}}{2}\\ -\frac{\sqrt{3}}{2} & \frac{1}{2}\end{bmatrix}$ |